



# NEW PRODUCTS

## TECHNOLOGY

### COMPONENTS & CHIPS

#### AI processors for business laptops

AMD has expanded its AI PC portfolio with the launch of the Ryzen PRO 8040 and 8000 series processors for business



laptops, mobile workstations, and desktops. The Ryzen PRO 8040 series offers x86 processors for mobile devices, while the

Ryzen PRO 8000 series introduces the first AI-enabled desktop processor, designed for high performance and power efficiency. These processors incorporate a CPU, GPU, and a dedicated neural processing unit (NPU), providing up to 16 dedicated NPU TOPS and 39 total system TOPS.

AMD

<https://www.amd.com/en.html>

#### Microcontrollers for AI-driven automotive applications

Infineon Technologies has introduced the AURIX TC4x microcontroller family, tailored to meet the growing needs of embedded AI in automotive safety-critical real-time applications. These



MCUs comply with AI-specific safety standards for SAFE AI, featuring an

ASIL-D certified AI-accelerator (PPU). They are ideal for developing AI-driven applications in motor control, battery management, vehicle motion control, and siren detection. Key features include up to six TriCore v1.8 processors, 25MB of on-chip Flash memory, high-speed communication interfaces, and support for robust AI integration under the SAFE AI framework, aligned

with ISO PAS 8800 standards.

Infineon Technologies AG

<https://www.infineon.com>

#### 32-bit microcontrollers for cybersecurity requirements

Microchip Technology Inc. has launched the PIC32CK family of 32-bit microcontrollers, featuring embedded security technologies to meet the enhanced cybersecurity mandates set



for 2024. These microcontrollers are integrated with a hardware security module (HSM) and an

ARM Cortex-M33 core equipped with TrustZone technology, designed to bolster security in digital applications ranging from consumer IoT to critical infrastructure. This integration facilitates compliance with new laws while reducing the complexity and cost of embedding advanced security in designs. They offer scalable security, memory, and connectivity options, including Ethernet, CAN FD, and USB.

Microchip Technology

<https://www.microchip.com>

#### Cost-effective and energy-efficient microcontrollers

STMicroelectronics has launched the STM32U0 microcontrollers, reducing energy usage by up to 50% compared to earlier models. The microcontrollers



are central in smart buildings and IoT applications, managing energy and resources through data collection and processing. These microcontrollers can be used in security cameras to activate

on motion, in smoke detectors, and in water metres for managing energy-sensitive functions. They include an LCD segment display controller that helps reduce PCB costs in devices like thermostats, retail labels, and factory systems, and come with on-chip oscillators to save space.

STMicroelectronics

<https://www.st.com>

#### Clock solution for enhanced high-speed systems

Renesas Electronics Corporation has expanded its timing solution offerings with the FemtoClock 3, an ultra-low jitter clock solution achieving 25fs-rms. Designed for wireline



infrastructure, data centres, and industrial uses, these devices are ideal for ad-

vanced high-speed interconnect systems, supporting diverse applications like telecom and data centre switches, medical imaging, and broadcast systems. Key features include integration of multiple frequency domains, low power usage, and compact, customisable options that comply with ITU-T standards, facilitating simpler, cost-effective clock tree architectures.

Renesas Electronics Corporation

<https://www.renesas.com/us/en>

#### CMOS global shutter sensors for advanced machine vision

OMNIVISION has launched two advanced CMOS global shutter image



sensors for machine vision applications, including industrial automation and robotics. The sensors feature the smallest-

ever 2.2-micron backside-illuminated